

STAT525 HOMEWORK#2

1. In four sentences or less, explain what a sampling distribution is and why it is important to know the sampling distribution of a test statistic.
2. Consider the Toluca Company example of Chapter 1. On page 46, a 95% confidence interval is constructed for the slope. Suppose instead the company were interested in the increase in mean work hours for an additional 6 units in the lot. Construct the 95% confidence interval for this quantity.
3. You fit the simple linear model to a data set and obtain estimates $b_0 = 2$, $b_1 = 5$, and $MSE = s^2 = 1.0$
 - (a) Suppose the data consisted of $n = 20$ cases and the standard error of b_1 , $s(b_1)$, is equal to 2. Construct the 95% CI for β_1 .
 - (b) Do you have statistical evidence X helps predict Y ? Explain.
 - (c) The 95% confidence interval for $E(Y)$ when $X = 5$ is $[23.38, 30.62]$. Find the 95% prediction interval when $X = 5$.
4. KNNL Problem 2.7
5. KNNL Problem 2.16
6. KNNL Problem 2.26. Use SAS to generate the ANOVA table and in place of part (c), generate the scatterplot and comment on the linearity.
7. Given that $R^2 = SSR/SSTO$, it can be shown that $R^2/(1 - R^2) = SSR/SSE$. If you have 25 cases and an $R^2 = 0.18$, what is the F statistic for the test that the slope is equal to zero?

1. In four sentences or less, explain what a sampling distribution is and why it is important to know the sampling distribution of a test statistic.

- A sampling distribution is the probability distribution of a statistic (e.g. sample mean) computed from repeated random samples of the same size from a population.
- A sampling distribution describes how the statistic varies due to random sampling.
- Knowing the sampling distribution of a test statistic allows us to determine probabilities, critical values, and p-values under the H_0 .
- This is essential for making valid statistical inferences about the population.

2. Consider the Toluca Company example of Chapter 1. On page 46, a 95% confidence interval is constructed for the slope. Suppose instead the company were interested in the increase in mean work hours for an additional 6 units in the lot. Construct the 95% confidence interval for this quantity.

For a 95 percent confidence coefficient, we require $t(.975; 23)$. From Table B.2 in Appendix B, we find $t(.975; 23) = 2.069$. The 95 percent confidence interval, by (2.15), then is:

$$3.5702 - 2.069(.3470) \leq \beta_1 \leq 3.5702 + 2.069(.3470)$$

$$2.85 \leq \beta_1 \leq 4.29$$

Thus, with confidence coefficient .95, we estimate that the mean number of work hours increases by somewhere between 2.85 and 4.29 hours for each additional unit in the lot.

Lot size $\uparrow$ by 6 units $\rightarrow$ mean work hours $\uparrow$ by: $6 \cdot \hat{\beta}_1$

$$CI = \hat{\beta}_1 \pm t_{\alpha/2, df} \cdot SE$$

$\Downarrow$ Additional 6 units for $\hat{\beta}_1$

$$\textcircled{1} \quad CI = (6) \hat{\beta}_1 \pm t_{\alpha/2, df} \cdot SE(6)$$

$$CI = (6)(3.5702) \pm 2.069 \cdot 0.3470(6)$$

$$CI = 21.4212 \pm 4.3077$$

$$(17.1135, 25.7289) \rightarrow (17.11, 25.73)$$

Lot size $\uparrow$ by 6 units

$$\textcircled{2} \quad (6) 2.85 \leq \hat{\beta}_1 \leq 4.29 (6)$$

$$\Downarrow$$

$$17.1 \leq 6 \hat{\beta}_1 \leq 25.74 \quad \checkmark$$

TABLE 2.1
Results for
Toluca
Company
Example
Obtained in
Chapter 1.

$n = 25$	$\bar{X} = 70.00$
$b_0 = 62.37$	$b_1 = 3.5702$
$\hat{Y} = 62.37 + 3.5702X$	$SSE = 54,825$
$\sum(X_i - \bar{X})^2 = 19,800$	$MSE = 2,384$
$\sum(Y_i - \hat{Y})^2 = 307,203$	

FIGURE 2.2
Portion of
MINITAB
Regression
Output—
Toluca
Company
Example.

The regression equation is
 $Y = 62.4 + 3.57 X$

Predictor	Coef	Stdev	t-ratio	p
Constant	62.37	26.18	2.38	0.026
X	3.5702	0.3470	10.29	0.000

$s = 48.82$ $R\text{-sq} = 82.2\%$ $R\text{-sq(adj)} = 81.4\%$

Analysis of Variance

SOURCE	DF	SS	MS	F	p
Regression	1	252378	252378	105.88	0.000
Error	23	54825	2384		
Total	24	307203			

$$\hat{\beta}_1 = 3.5702$$

$$t_{0.975, 23} = 2.069$$

$$SE = 0.3470$$

3. You fit the simple linear model to a data set and obtain estimates $b_0 = 2$, $b_1 = 5$, and $MSE = s^2 = 1.0$

(a) Suppose the data consisted of $n = 20$ cases and the standard error of b_1 , $s(b_1)$, is equal to 2. Construct the 95% CI for β_1 .

$$CI = \hat{\beta}_1 \pm t_{\alpha/2, df} \cdot SE$$

$$\begin{aligned} \circ \hat{\beta}_1 &= 5 \\ \circ df &= 20 - 2 = 18 \\ \circ t_{0.975, 18} &= 2.101 \\ \circ s(b_1) &= 2 \end{aligned}$$

$$CI = 5 \pm 2.101 \cdot 2$$

$$(0.798, 9.202) \rightarrow (0.80, 9.20)$$

95% CI for β_1

(b) Do you have statistical evidence X helps predict Y ? Explain.

① $(0.80, 9.20)$, the 95% CI for β_1 , does not contain 0, meaning we reject H_0 .

② $t = \hat{\beta}_1 / SE(\hat{\beta}_1) = 5/2 = 2.5 > t_{0.975, 18} = 2.101$, meaning we reject H_0 .

Hence, there is statistical significance that the slope is different from 0.

Therefore X helps predict Y .

(c) The 95% confidence interval for $E(Y)$ when $X = 5$ is $[23.38, 30.62]$. Find the 95% prediction interval when $X = 5$.

$$\text{Prediction Interval} = \hat{Y} \pm t_{\alpha/2, df} \cdot S \cdot \sqrt{1+h}$$

$$\text{Confidence Interval} = \hat{Y} \pm t_{\alpha/2, df} \cdot S \cdot \sqrt{h}$$

$$\begin{aligned} \circ Y &= 5X + 2 \\ \circ \hat{Y}_5 &= 5(5) + 2 = 27 \end{aligned}$$

$$\begin{aligned} CI \text{ Higher} &= 27 + 2.101 \cdot 1 \cdot \sqrt{h} = 30.62 \\ \sqrt{h} &= 1.723 \\ h &= 2.9687 \checkmark \end{aligned}$$

$$PI \text{ higher} = 27 + 2.101 \cdot 1 \cdot \sqrt{1+2.9687} = 31.1855$$

$$PI \text{ lower} = 27 - 2.101 \cdot 1 \cdot \sqrt{1+2.9687} = 22.8145$$

$$\begin{aligned} CI \text{ Lower} &= 27 - 2.101 \cdot 1 \cdot \sqrt{h} = 23.38 \\ \sqrt{h} &= 1.723 \\ h &= 2.9687 \checkmark \end{aligned}$$

↓

95% Prediction Interval when $X=5$:
(22.81, 31.19)

4. KNNL Problem 2.7

2.7 Refer to **Plastic hardness** Problem 1.22.

- a. Estimate the change in the mean hardness when the elapsed time increases by one hour. Use a 99 percent confidence interval. Interpret your interval estimate.

```

1 data hardness;
2 input hardness time;
3 datalines;
4 199 16
5 205 16
6 196 16
7 200 16
8 218 24
9 220 24
10 215 24
11 223 24
12 237 32
13 234 32
14 235 32
15 230 32
16 250 40
17 248 40
18 253 40
19 246 40
20 ;
21 run;
22
23 proc reg data=hardness;
24     model hardness = time / clb alpha=0.01;
25 run;

```

The REG Procedure
Model: MODEL1
Dependent Variable: hardness

Number of Observations Read	16
Number of Observations Used	16

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	5297.51250	5297.51250	506.51	<.0001
Error	14	146.42500	10.45893		
Corrected Total	15	5443.93750			

Root MSE	3.23403	R-Square	0.9731
Dependent Mean	225.56250	Adj R-Sq	0.9712
Coeff Var	1.43376		

Parameter Estimates						
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t	99% Confidence Limits
Intercept	1	168.60000	2.65702	63.45	<.0001	160.69046 176.50954
time	1	2.03438	0.09039	22.51	<.0001	1.76529 2.30346

• $\hat{\beta}_1 = 2.03438$

• CI for $\hat{\beta}_1$: (1.76529, 2.30346)

- 2.034 units per hour in estimated change in mean hardness for every 1 hour ↑.
- With 99% confidence, we estimate that for every one-hour ↑ in elapsed time, the mean plastic hardness ↑ by between 1.77 to 2.30 units.
- CI does not contain 0, we reject H_0 , there is statistical evidence that ↑ elapsed time, ↑ plastic hardness.

- b. The plastic manufacturer has stated that the mean hardness should increase by 2 Brinell units per hour. Conduct a two-sided test to decide whether this standard is being satisfied; use $\alpha = .01$. State the alternatives, decision rule, and conclusion. What is the P -value of the test?

$$H_0: \hat{\beta}_1 = 2$$

$$H_A: \hat{\beta}_1 \neq 2$$

$$t = \frac{\hat{\beta}_1 - \beta_{H_0}}{SE(\hat{\beta}_1)}$$

$$t = \frac{2.03438 - 2}{0.09039} = 0.380352 < t_{0.995, 14} = 2.624$$

p-value: 0.7071

↓

- Fail to reject H_0 at the $\alpha = 0.01$ significance level.
- There is no statistical evidence that the actual mean ↑ in hardness per hour is different from the manufacturer's standard of 2 Brinell units.

- c. Obtain the power of your test in part (b) if the standard actually is being exceeded by .3 Brinell units per hour. Assume $\sigma\{b_1\} = .1$. SE

- $H_0: \beta_1 = 2$
- $H_A: \beta_1 \neq 2$
- True $\beta_1 = 2.3$
- $\sigma(\hat{\beta}_1) = 0.1$
- $\alpha = 0.01$ (two-sided)
- $df = 19$
- $t_{0.995, 19} = 2.624$

Step 1. Compute Noncentrality Parameter

$$\delta = \frac{\beta_{true} - \beta_0}{\sigma(\hat{\beta}_1)}$$

$$\delta = \frac{2.3 - 2.0}{0.1} = 3$$

Step 3. Compute Power

$$\text{Power} = P(T < -2.977 | \delta = 3) + P(T > 2.977 | \delta = 3)$$

$$\text{Power} = 0.530 = 53\%$$

Step 2. Critical t-value

- $\alpha = 0.01$
- $df = 19$

$$t_{0.005, 19} \approx \pm 2.977$$

• There is a 53% power to detect true slope of 2.3 against $H_0 = 2$ at $\alpha = 0.01$ level.

5. KNNL Problem 2.16

2.16. Refer to Plastic hardness Problem 1.22.

- a. Obtain a 98 percent confidence interval for the mean hardness of molded items with an elapsed time of 30 hours. Interpret your confidence interval.

```

20 . 30
21 ;
22 run;

24 proc reg data=hardness;
25     model hardness = time / (clm) alpha=0.02;
26     id time;
27 run;

```

(227.4569, 231.8056)

- With 98% confidence, the mean hardness of molded items with an elapsed time of $X = 30$ hours is between 227.46 to 231.81 Brinell units.

Refer to SAS output from Q4 (2.7)

The REG Procedure
Model: MODEL1
Dependent Variable: hardness

Output Statistics a)

Obs	time	Dependent Variable	Predicted Value	Std Error Mean Predict	98% CL Mean	Residual
1	16	199	201.1500	1.3529	197.5993 204.7007	-2.1500
2	16	205	201.1500	1.3529	197.5993 204.7007	3.8500
3	16	196	201.1500	1.3529	197.5993 204.7007	-5.1500
4	16	200	201.1500	1.3529	197.5993 204.7007	-1.1500
5	24	218	217.4250	0.8857	215.1006 219.7494	0.5750
6	24	220	217.4250	0.8857	215.1006 219.7494	2.5750
7	24	215	217.4250	0.8857	215.1006 219.7494	-2.4250
8	24	223	217.4250	0.8857	215.1006 219.7494	5.5750
9	32	237	233.7000	0.8857	231.3756 236.0244	3.3000
10	32	234	233.7000	0.8857	231.3756 236.0244	0.3000
11	32	235	233.7000	0.8857	231.3756 236.0244	1.3000
12	32	230	233.7000	0.8857	231.3756 236.0244	-3.7000
13	40	250	249.9750	1.3529	246.4243 253.5257	0.0250
14	40	248	249.9750	1.3529	246.4243 253.5257	-1.9750
15	40	253	249.9750	1.3529	246.4243 253.5257	3.0250
16	40	246	249.9750	1.3529	246.4243 253.5257	-3.9750
17	30	.	229.6313	0.8285	227.4569 231.8056	.

Sum of Residuals	0
Sum of Squared Residuals	146.42500
Predicted Residual SS (PRESS)	194.01315

$$\hat{y} = 2.03438x + 168.60$$

$$\hat{y}_{30} = 2.03438(30) + 168.60 = 229.631 \checkmark$$

- b. Obtain a 98 percent prediction interval for the hardness of a newly molded test item with an elapsed time of 30 hours.

```

24 proc reg data=hardness;
25     model hardness = time / cli alpha=0.02;
26     id time;
27 run;

```

(220.8695, 238.3930)

- With 98% confidence, the new hardness of molded items with an elapsed time of $X=30$ hours is between 220.87 to 238.39 Brinell units.

The REG Procedure
Model: MODEL1
Dependent Variable: hardness

Output Statistics							
Obs	time	Dependent Variable	Predicted Value	Std Error Mean Predict	98% CL Predict		Residual
1	16	199	201.1500	1.3529	191.9496	210.3504	-2.1500
2	16	205	201.1500	1.3529	191.9496	210.3504	3.8500
3	16	196	201.1500	1.3529	191.9496	210.3504	-5.1500
4	16	200	201.1500	1.3529	191.9496	210.3504	-1.1500
5	24	218	217.4250	0.8857	208.6248	226.2252	0.5750
6	24	220	217.4250	0.8857	208.6248	226.2252	2.5750
7	24	215	217.4250	0.8857	208.6248	226.2252	-2.4250
8	24	223	217.4250	0.8857	208.6248	226.2252	5.5750
9	32	237	233.7000	0.8857	224.8998	242.5002	3.3000
10	32	234	233.7000	0.8857	224.8998	242.5002	0.3000
11	32	235	233.7000	0.8857	224.8998	242.5002	1.3000
12	32	230	233.7000	0.8857	224.8998	242.5002	-3.7000
13	40	250	249.9750	1.3529	240.7746	259.1754	0.0250
14	40	248	249.9750	1.3529	240.7746	259.1754	-1.9750
15	40	253	249.9750	1.3529	240.7746	259.1754	3.0250
16	40	246	249.9750	1.3529	240.7746	259.1754	-3.9750
17	30	.	229.6313	0.8285	220.8695	238.3930	.

- c. Obtain a 98 percent prediction interval for the mean hardness of 10 newly molded test items, each with an elapsed time of 30 hours.

$$SE_{adj} = \sqrt{SE + (MSE/m)}$$

$$\text{Prediction Interval} = \hat{y} \pm t_{\alpha/2, df} \cdot SE_{adj}$$

Refer to SAS output from Q4 (2.7)

- $MSE = 10.45893$

- $m = 10$

- $SE(\hat{y}_{30}) = 0.8285$

- $t_{0.01, 19} = 2.6245$

$$SE_{adj} = \sqrt{(0.8285)^2 + (10.45893/10)} = 1.31617$$

$$\text{Prediction Interval} = 229.631 \pm 2.6245(1.31617)$$

$$\Downarrow$$

$$(226.177, 233.085)$$

- With 98% confidence, the 10 new hardness of molded items with an elapsed time of $X=30$ hours is between 226.18 to 233.09 Brinell units.

- d. Is the prediction interval in part (c) narrower than the one in part (b)? Should it be?

Yes. Interval in (c) is narrower in (d), as it should be.

- Averaging over 10 items $\downarrow$ variability, $\downarrow$ SE, hence $\downarrow$ interval.

WT Band

- e. Determine the boundary values of the 98 percent confidence band for the regression line when $X_h = 30$. Is your confidence band wider at this point than the confidence interval in part (a)? Should it be?

```

26 proc reg data=hardness;
27     model hardness = time / clm cli alpha=0.02; /* 98% CI and PI */
28     output out=a2 p=predicted stdp=stdp uclm=uclm lclm=lclm ucl=ucl lcl=lcl;
29     id time;
30 run;
31
32 data a3;
33     set a2;
34     /* df = n - 2 = 16 - 2 = 14 */
35     whl = predicted - sqrt(2 * finv(1 - 0.02, 2, 14)) * stdp;
36     whu = predicted + sqrt(2 * finv(1 - 0.02, 2, 14)) * stdp;
37 run;
38
39 proc print data=a3;
40     where time = 30;
41     var time predicted whl whu lclm uclm lcl ucl;
42 run;

```

Obs	time	predicted	whl	whu	lclm	uclm	lcl	ucl
17	30	229.631	226.949	232.313	227.457	231.806	220.869	238.393

(a):CI

(b):PI

(226.95, 232.31)
Working-Hotelling Band
Confidence Band

- Yes, WT Band is slightly wider than regular CI for the mean, as it should be.
- WT accounts for simultaneous inference across all values of X , not just at $X=30$.
- WT is more conservative than CI, which is why it is wider.

6. KNNL Problem 2.26. Use SAS to generate the ANOVA table and in place of part (c), generate the scatterplot and comment on the linearity.

2.26. Refer to **Plastic hardness** Problem 1.22.

Refer to SAS output from Q4 (2.7)

a. Set up the ANOVA table.

```

23 proc reg data=hardness;
24     model hardness = time;
25 run;

```

The REG Procedure
Model: MODEL1
Dependent Variable: hardness

Number of Observations Read	16
Number of Observations Used	16

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	5297.51250	5297.51250	506.51	<.0001
Error	14	146.42500	10.45893		
Corrected Total	15	5443.93750			

- b. Test by means of an F test whether or not there is a linear association between the hardness of the plastic and the elapsed time. Use $\alpha = .01$. State the alternatives, decision rule, and conclusion.

$$H_0: \beta_1 = 0 \text{ (No Linear Trend)}$$

$$H_A: \beta_1 \neq 0 \text{ (Linear Trend)}$$

Refer to SAS output from Q4 (2.7)

• F-statistic : 506.51

• p-value : $< 0.0001 < 0.01 = \alpha$

↓

Reject H_0

• There is strong evidence of linear relationship between elapsed time and plastic hardness.

• Time is a statistically significant predictor of hardness at 1% level.

```
23 proc reg data=hardness;
24     model hardness = time / alpha=0.01;
25 run;
```

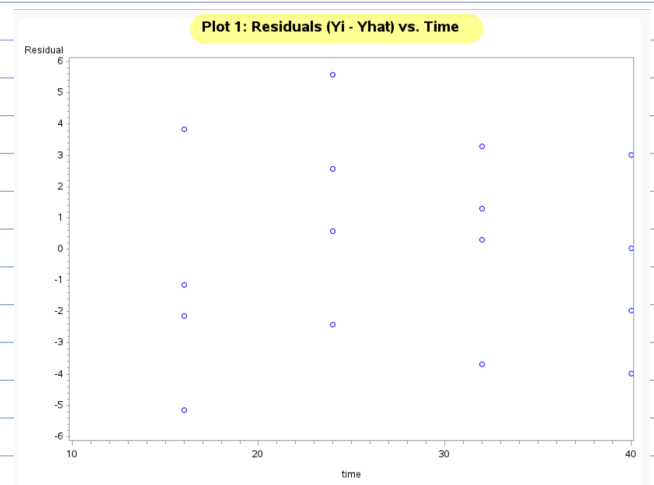
- c. Plot the deviations $Y_i - \hat{Y}_i$ against X_i on a graph. Plot the deviations $\hat{Y}_i - \bar{Y}$ against X_i on another graph, using the same scales as for the first graph. From your two graphs, does SSE or SSR appear to be the larger component of SSTO? What does this imply about the magnitude of R^2 ?

$$e = Y_i - \hat{Y}_i = \text{Residuals}$$

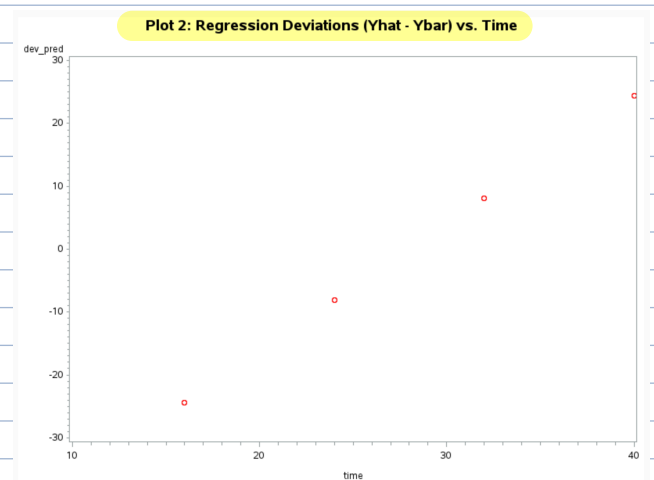
$$\hat{Y}_i - \bar{Y} = \text{Deviation from the mean}$$

```
24 proc reg data=hardness;
25     model hardness = time;
26     output out=regout p=pred r=resid; /* Save Yhat and residuals */
27 run;
28
29 proc means data=regout mean noprint;
30     var hardness;
31     output out=meanout mean=Ybar;
32 run;
33
34 data regout2;
35     if _n_ = 1 then set meanout;
36     set regout;
37     dev_pred = pred - Ybar; /* SSR component */
38 run;
```

```
40 /* Plot Residuals vs Time (Yi - Yhat) */
41 title "Plot 1: Residuals (Yi - Yhat) vs. Time";
42 symbol1 v=circle c=blue i=none;
43 proc gplot data=regout2;
44     plot resid*time;
45     axis1 label=("Elapsed Time");
46     axis2 label=("Residual");
47 run;
48 quit;
49
50 /* Plot Deviations (Yhat - Ybar) vs Time */
51 title "Plot 2: Regression Deviations (Yhat - Ybar) vs. Time";
52 symbol1 v=circle c=red i=none;
53 proc gplot data=regout2;
54     plot dev_pred*time;
55     axis1 label=("Elapsed Time");
56     axis2 label=("Deviation from Mean");
57 run;
```



random scatter → good.



linear trend → good.

$$SSR = 5297.5125 \gg 146.4250 = SSE$$

• SSR is much larger than SSE

$$R^2 = 5297.5125 / 5443.9375 = 0.9731$$

• This high value (97.31%) confirms that most of total variation, (SSTO), is explained by the regression (SSR), not by residuals (SSE).

↓ Linearity Assumption is well satisfied ↑


```

59 proc sgplot data=hardness;
60     scatter x=time y=hardness;
61     reg x=time y=hardness; /* Adds fitted regression line */
62     title "Scatterplot of Hardness vs. Time with Regression Line";
63 run;

```

- $\hat{Y}$ v.s. X shows points follow a straight-line pattern along the fitted regression line with no curvature nor systematic deviation.
- Linear model is appropriate.



d. Calculate R^2 and r .

linear trend $\rightarrow$ good.

$$R^2 = SSR/SSTO$$

$$r = \sqrt{R^2}$$

$$R^2 = 5297.5125 / 5443.9375$$

$$r = \sqrt{0.9731}$$

$$R^2 = 0.9731$$

$$r = 0.9865$$

• Since $\hat{B}_1$ is positive, correlation, r , is also positive.

7. Given that $R^2 = SSR/SSTO$, it can be shown that $R^2/(1 - R^2) = SSR/SSE$. If you have 25 cases and an $R^2 = 0.18$, what is the F statistic for the test that the slope is equal to zero?

$$F = \frac{MSR}{MSE} = \frac{SSR/1}{SSE/(n-2)}$$

$$F = \frac{SSR(n-2)}{SSE}$$

$\Downarrow$ plug in $R^2/(1-R^2) = SSR/SSE$

$$F = (R^2/1-R^2) \cdot (n-2)$$

$$F = (0.18/1-0.18) \cdot (25-2) = 5.04878$$